Mathematics for Computing

Assignment 1 : Set Theory

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Assignment No.1

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Introduction

- **What is a set?**

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  - Example of a set: Empty set (no items): $\{ \}$ or ϕ

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  - Example of a set: Empty set (no items): $\{\ \}$ or ϕ

  - Example of not a set: Set of all tall people in this room
  - Example of not a set: The collection of the best football players in the world.

Set Builder Form

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   $$Q_1 = \{(x, y) \mid x, y \in \mathbb{R} \text{ and } x > 0 \text{ and } y > 0\}$$

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$$S = \{x \mid x^2 < 50\ \forall\ x \in \mathbb{Z}\}$$

Example of Sets (from A.I. and M.L. Domain)

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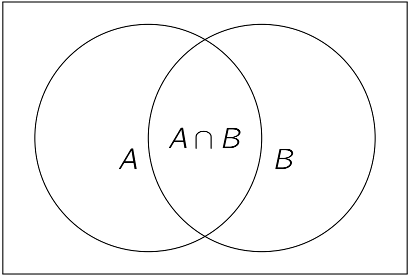
Inclusion - Exclusion Principle

- **What is Inclusion Exclusion Principle?**
  The Inclusion Exclusion principle is a counting method used to find the number of elements in the union of multiple sets.
- For two sets **A and B**

$$|A \cup B| = |A| + |B| - |A \cap B|$$

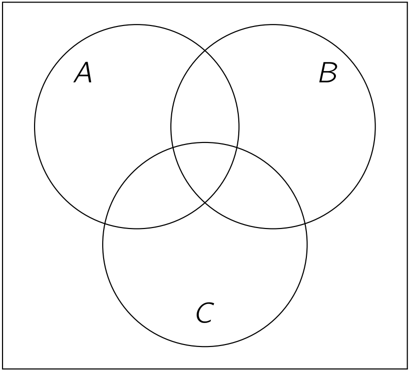


$$|A \cup B|$$

Inclusion - Exclusion Principle

- For three sets **A, B and C**

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



$$|A \cup B \cup C|$$

General form of Inclusion-Exclusion Principle

For finite sets $A_1, A_2, A_3, \ldots, A_n$, the total number of elements in their union is:

$$\begin{aligned}\left|\bigcup_{i=1}^{n} A_i\right| &= \sum_{i=1}^{n} |A_i| \\ &- \sum_{1\le i<j\le n} |A_i \cap A_j| \\ &+ \sum_{1\le i<j<k\le n} |A_i \cap A_j \cap A_k| \\ &- \cdots \\ &+ (-1)^{n-1}|A_1 \cap A_2 \cap \cdots \cap A_n|\end{aligned}$$

General form of Inclusion-Exclusion Principle

For finite sets $A_1, A_2, A_3, \ldots, A_n$, the total number of elements in their union is:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{k=1}^{n} (-1)^{k-1} \left(\sum_{1 \le i_1 < i_2 < \cdots < i_k \le n} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| \right)$$

What is a Poset?

A partially ordered set is a a set where we have defined a specific rule fro how elements in a set relate to each other.

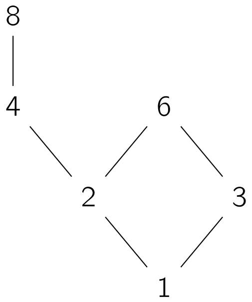
For a set to be a poset, the following three conditions must be met

- Reflexive: Every item relates to itself.
- Antisymmetric: If item **A** relates to item **B** and item **B** relates back to item **A**, then they must actaually be the exact same item.
- Transitive If **A** relates to **B** and **B** relates to **C** , then **A** automatically relates to **C**

Hasse Diagram

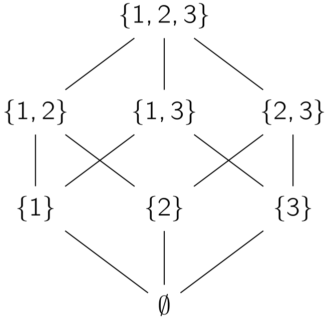
A graphical representation of a partial order relation in which all arrowheads are understood to be pointing upward is known as the Hasse Diagram.

Example: Let $A = \{1, 2, 3, 4, 6, 8\}$ be ordered by the relation "a divides b".



Hasse Diagram

Example: The power set of $S = \{1, 2, 3\}$ ordered by subset inclusion ($\subseteq$).



Thank You

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